DE LA SALLE HOLY CROSS COLLEGE

Grade 10 **Date: November 2021**

Physical Science **Total: 130 marks**

Memorandum Paper 1 **Time: 120 minutes**

Examiner: A. Kamedza **Moderator: K. Tam**

Question 1 Multiple Choice

1.1 C ✓✓
1.2 D ✓✓
1.3 D ✓✓
1.4 B ✓✓
1.5 D ✓✓
1.6 C ✓✓
1.7 C ✓✓
1.8 D ✓✓
1.9 C ✓✓
1.10 C ✓✓

[10 x 2 = 20]

Question 2

2.1.1 Longitudinal ✓ (1)

2.1.2 It is a wave where the disturbance(particles) of the medium is parallel to the direction of propagation of the wave ✓✓ (2)

2.1.3 0,45 mm✓ (2)

2.1.4 1 wavelength passes in 8 s ✓

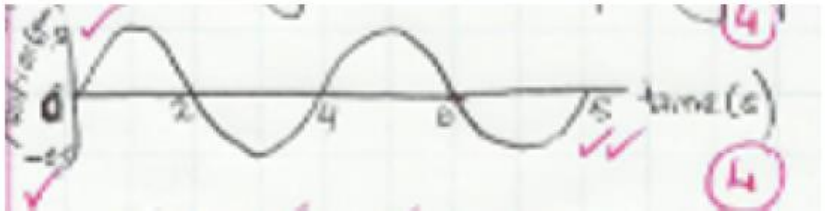
f = $\frac{1}{8}$✓ = 0,125 Hz✓ (3)

2.1.5 T = $\frac{1}{f} = \frac{1}{0,125}$✓ = 8 s✓ or read off graph as 8 s✓✓ **no c.o.e** (2)

2.1.6 v = fλ (3)

$\lambda = \frac{0,15✓}{0,125✓}$ = 1,2 m✓

2.1.7 (4)



2.1.8 Points in phase A & B✓ (1)

2.2.1 v = fλ (3)

$\lambda = \frac{3\times10^{8}✓}{24\times10^{9}✓}$ = 0,0125 m✓

2.2.2 The temperature of the ball will increase ✓. (1)

2.2.3 The difference in frequency is directly✓proportional to the speed of the ball✓(2)

2.2.4 20✓ m·s^{-1} ✓ (2)

[25 marks]

Question 3

3.1 Electrons are transferred from the rod to the cloth✓. The rod has less electrons than protons and is therefore positively charged. ✓ (2)

3.2 The metal paint has delocalized/free-moving electrons✓. When the rod is brought near the pith ball, the electrons in the ball move to one side of the ball nearest to the positive rod✓. (2)

3.3 The pith ball is polarized when the rod is brought near it✓. The negative side of the ball is attracted to the positive rod✓ (2)

3.4 The pith ball will touch the rod(& share the charges equally between the 2 objects) and will then spring/move away✓✓ (2)

3.5 When the rod and ball touch, they share their charges equally✓ between them resulting in them both having a positive charge(same charge) and repelling✓ (2)

[10 marks]

Question 4

4.1 Magnetic field- a region in space✓ where a magnet or magnetic material will experience a magnetic force ✓. (2)

4.2 - never touch/cross✓ (2)
- move from North to South✓
- imaginary, 3-dimensional lines✓ (any 2)
- the closer the field lines the stronger the field

4.3 B & C are like/similar poles✓; both South poles✓; equal strength.✓**(2 only)** (2)

4.4 C to D✓ lines inside magnet point from South to North✓ so that outside is North to south✓ (3)

4.5 Repulsion is stronger✓ the closer the magnets are to each other. ✓ (2)

4.6 Animals use it for navigation during migration✓ (1)

[12 marks]

Question 5

5.1 V_{pd} – is the work done per unit positive charge✓. (1)

5.2.1 10 V✓
5.2.2 0 A ✓. (2)

5.3 $\frac{1}{R} = \frac{1}{4} + \frac{1}{2} = \frac{3}{4}$✓ (3)
RT = $\frac{4}{3}$ + 5 + 6✓ =12,33 Ω✓

5.4 I = $\frac{V}{R} = \frac{10}{12,33}$✓= 0,81 A✓ (2)

5.5 V = IR = 0,81✓ x 1,33✓ = 1,08 V ✓ (3)

5.6 V = IR = 0,81 x 6 ✓ = 4,86 V ✓ (2)

5.7 I = $\frac{V}{R} = \frac{1,08✓}{4✓}$ = 0,27 A✓ (3)

5.8 Bigger✓ the lower the resistance the larger the current✓ (2)

[18 marks]

QUESTION 6

6.1.1 $\Delta x = 6+2+3+1+1$✓$= 13$ m. ✓ (2)

6.1.2 Displacement is the change in position relative to a reference/starting point✓
$\Delta x = 5$ m✓ at a bearing of 53,1°✓ (3)

6.1.3 Average speed = $\frac{distance\ travelled}{time\ taken}$✓ $= \frac{13}{30}$✓ $= 0{,}43$ m·s^{-1} ✓ (3)

6.1.4 Average speed = $\frac{displacement}{time\ taken}$✓ $= \frac{5}{30}$✓ $= 0{,}17$ m·s^{-1} ✓ (3)

6.2.1 Acceleration- the rate of change of velocity✓✓ (2)

6.2.2 The car is moving South ✓, slowing down at a constant rate✓/with constant deceleration. (2)

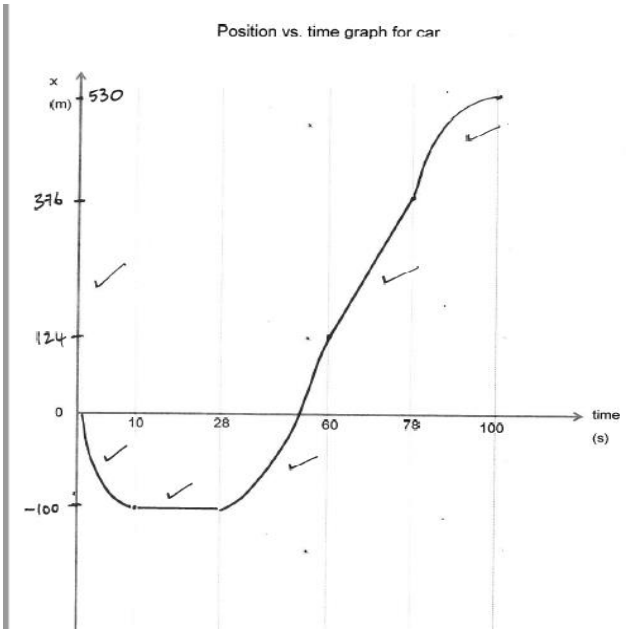
6.2.3 $a = \frac{\Delta v}{\Delta t}$ ✓$= \frac{0-14}{100-78}$✓ $= -0{,}636 = 0{,}636$ m·s^{-2} South ✓ (3)

6.2.4 Δx = Area under graph✓ = ½(10)(-20) + (18x14) ✓ + ½(32+22)(14) ✓
= 530 m North✓ (4)

6.2.5 $v_{ave} = \frac{\Delta x}{\Delta t} = \frac{124}{60}$✓✓ $= 2{,}07$ m·s^{-1} ✓ (3)

6.2.6 B/C✓ (1)

6.2.7 (6)



6.3.1 Em – the sum of an object's gravitational potential energy and kinetic energy. ✓ (1)

6.3.2 It is the same✓. Mechanical energy is conserved✓ in an isolated, frictionless system. (2)

6.3.3 $E_m = mgh + ½ mv^2$ ✓= (500)(9,8)(10) + ½ (500)(6^2) ✓ = 58 000J✓ (3)

6.3.4 $E_m = mgh + ½ mv^2$ (4)
58000✓ = (500)(9,8)(5) ✓ + ½ (500)(v^2) ✓
v =11,58 $m{\cdot}s^{-1}$ ✓

6.3.5 Ep = mgh = (500)(9,8)(20) ✓ = 98 000J✓ (3)
Ep>Em, therefore cart will not reach point C✓

[45 marks]

Total: 130 marks